Homework:

What is the most probable 4 day weather sequence? Assume weather={**S**unny, Cloudy , **R**aining} with transition probability matrix:

$$A = \begin{pmatrix} .8 & .15 & .05 \\ .38 & .6 & .02 \\ .75 & .05 & .2 \end{pmatrix}$$

4-day sequence can start any day

W1 W2 W3 W4 W5 W6 W7 W8 W9 W10 W11 W12 … … …

Hint: today’s distribution is the stationary distribution

Homework

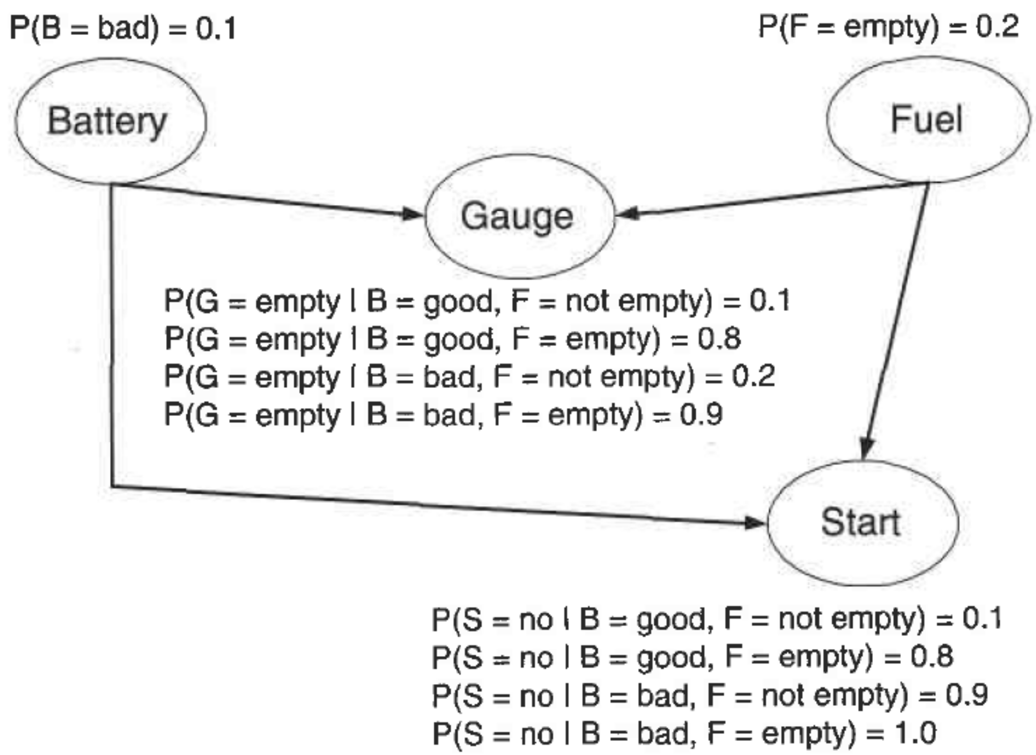
If today is rainny, will tommorrow be sunny? Or the day after tommorrow be sunny ?

If today is rainny, how many days I need to wait to get a sunny day?

$$A = \begin{pmatrix} .8 & .15 & .05 \\ .38 & .6 & .02 \\ .75 & .05 & .2 \end{pmatrix}$$

Some possible sequences:

Homework PR3.



Tasks:
1. Write down the probability distribution decomposition.
2. Compute:

(a) $P(\mathrm{B} = \mathrm{good},\ \mathrm{F} = \mathrm{empty},\ \mathrm{G} = \mathrm{empty},\ \mathrm{S} = \mathrm{yes})$.

(b) $P(\mathrm{B} = \mathrm{bad},\ \mathrm{F} = \mathrm{empty},\ \mathrm{G} = \mathrm{not\ empty},\ \mathrm{S} = \mathrm{no})$.

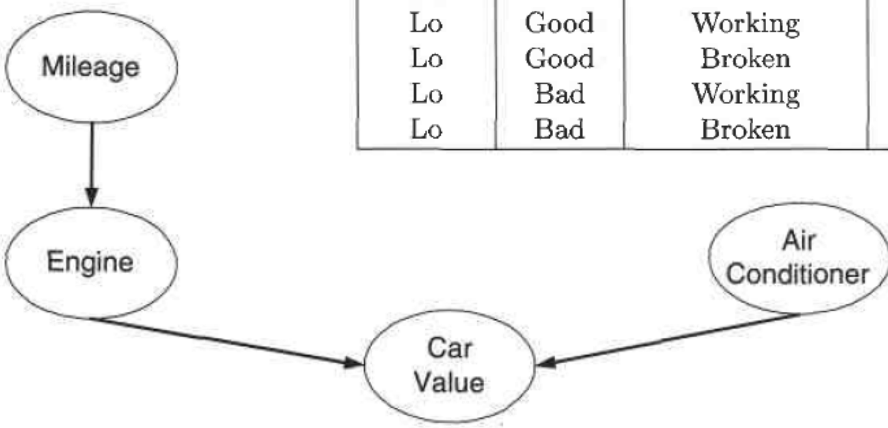
(c) Given that the battery is bad, compute the probability that the car will start.

Homework: PR3

M E A Car Value=C

Mileage	Engine	Air Conditioner	Number of Records with Car Value=Hi	Number of Records with Car Value=Lo
Hi	Good	Working	3	4
Hi	Good	Broken	1	2
Hi	Bad	Working	1	5
Hi	Bad	Broken	0	4
Lo	Good	Working	9	0
Lo	Good	Broken	5	1
Lo	Bad	Working	1	2
Lo	Bad	Broken	0	2

This line means there are 3+4 data instances:
3 instances for car-value= High,
4 instances for car-value = Low



Given Bayesian Network, and the training data instances.
Tasks:
1. Write down the probability distribution decomposition.
2. Learn network parameters: P(M), P(A), P(E|M), P(C|E,A).
   Hint: When estimate P(C|E,A), you need to integrate out *M.*
   Thus there are 3+4+9+0 data instances with E=good AND A=working ,
   3+9 instances for C=Hi, and
   - 4+0 instances for C=low.
3. Compute Prob(Engine = bad, Air Condition = Broken)